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Psychological Assessment and Intervention in Rehabilitation

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SUMMARY PEARLS

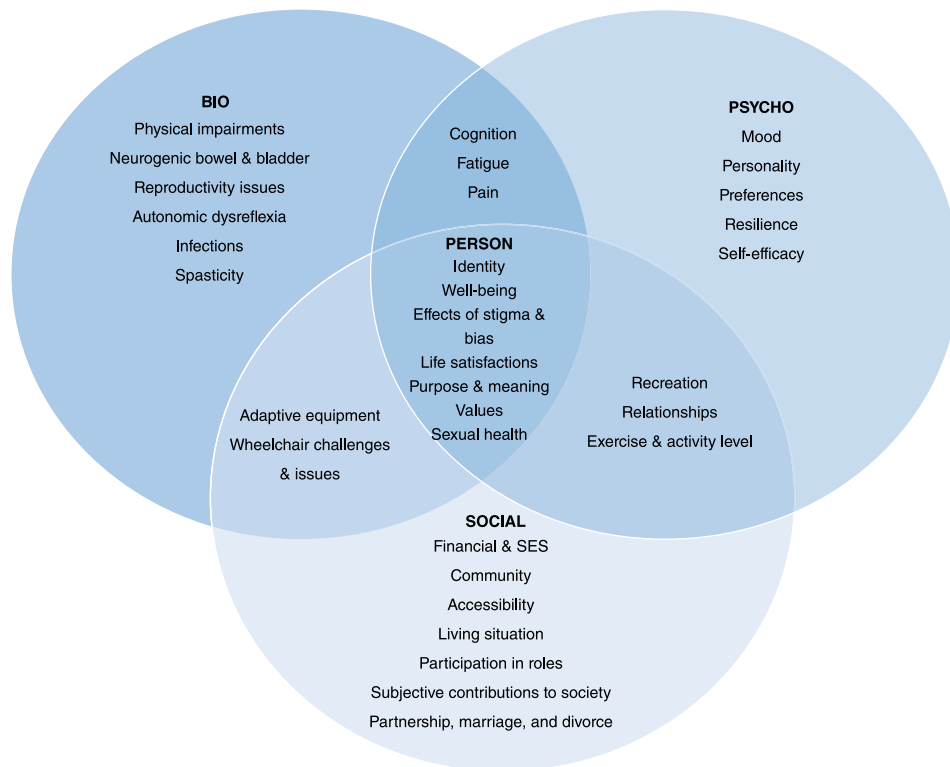
- Disability or illness creates psychological challenges for individuals that can put them at risk for psychopathology, including mood disorders, posttraumatic stress disorder, substance abuse, and increased suicide risk that can impact rehabilitation outcomes.
- A biopsychosocial approach is needed to understand and support the individual's adjustment process and address the emotional, psychological, and social dimensions involved in a patient-centered approach to rehabilitation care.
- Most individuals adjust to their new circumstances by utilizing adaptive coping strategies, finding meaning and purpose in remaining abilities, and maintaining a sense of hope that allows them enhanced resilience and quality of life.
- Maladjustment can manifest in a variety of ways over the course of recovery requiring routine thoughtful assessment and referrals to mental health resources for assistance in addressing issues interfering with rehabilitation and recovery.
- As most individuals rely on family systems in the community, it is important for the rehabilitation team to address the family's informational, emotional, instrumental, and training needs to encourage healthy caregiver coping and adjustment.
- Rehabilitation specialists can influence adjustment by monitoring individuals' recoveries in a comprehensive, multifaceted approach to aid what ultimately will be a lifelong process involving ongoing adaptation to dynamic circumstances, redefining identity, maintaining social connection, and finding new sources of meaning and fulfillment.

Background

As specialists in physical medicine and rehabilitation (PM&R), physiatrists encounter patients facing a multitude of medical conditions, from temporary to chronic, that bring unique challenges. These conditions may involve (1) extended or repeated medical procedures and hospitalizations, (2) pain, (3) disruptions in sleep, energy, and general well-being, and (4) changes in functional capacities and physical appearance. The psychological and emotional impact of these conditions often disrupts individuals' identity, as they struggle to redefine themselves in the setting of new limitations and capabilities. Roles within family, work, and social spheres must be reevaluated, challenging established norms and expectations. The loss of occupational roles can lead to a reassessment of one's purpose and self-worth. Injury, illness, and acquired disability can significantly alter an individual's ability to engage in essential roles and activities at home, work, and in the community, as well as their overall life trajectory, dramatically altering their sense of self and their relationships with others and the world around them.⁶ Social support is crucial for psychological resilience and flourishing, whereas a lack of support can exacerbate feelings of isolation and despair. The quality and stability of social relationships positively influence adjustment and

self-esteem, whereas social stigma and discrimination can undermine self-worth and hinder societal participation. The biopsychosocial model, formulated by George Engel in 1977, acknowledges that reactions to health and illness are influenced by the interaction of biologic, psychological, and social factors. As an example, [Fig. 5.1](#) provides a visual summary of the biopsychosocial consequences that can follow a spinal cord injury (SCI).

As an example, SCI has biologic consequences, such as impaired sensation and mobility, increased spasticity, neurogenic bowel and bladder functioning, autonomic dysreflexia, increased risk for infection, fatigue, and pain. Psychological consequences can include changes in mood, personality, self-image, and sense of self-efficacy. Social consequences can include changes in financial and residential status; accessibility issues limiting participation in social and community activities; inability to participate in essential social, family, and community roles; exposure to disability bias; and discrimination. These biologic, psychological, and social consequences can combine to impact individuals' sense of identity, well-being, purpose, values, meaning, and life satisfaction, which can potentially have a negative impact on their mental health. Patients facing acute or chronic conditions undergo a complex adjustment process within this framework.⁷¹ Appreciating the psychosocial impact of acute or chronic conditions is an

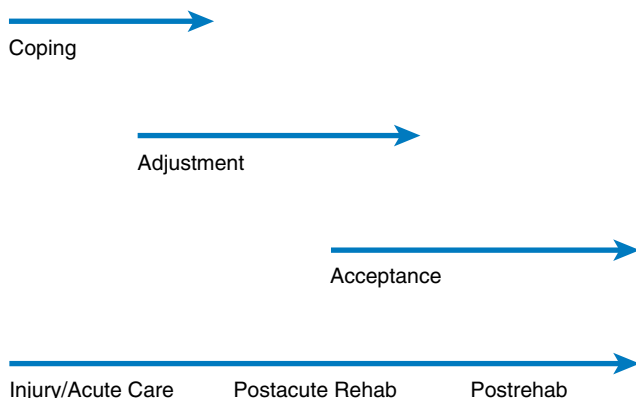


• **Fig. 5.1** Biopsychosocial model of spinal cord injury. SES, Socioeconomic status. (From Budd MA, Gater Jr DR, Channell I. Psychosocial consequences of spinal cord injury: a narrative review. *J Pers Med*. 2022;12(7):1178.)

essential component of patient-centered care, allowing health care providers to support patients in moving through this adjustment process, promoting resilience and enhancing their psychological well-being and quality of life.

Stages of Recovery

As Fig. 5.2 suggests, patients in rehabilitation typically progress through stages of acceptance over the course of their recovery beginning in the initial acute care and continuing through postrehabilitation integration into the community. Though the timeline and trajectory can vary widely, these phases provide a framework for understanding the adjustment process and the varied trajectories and timelines individuals may experience.³⁷ It is highly individualized and influenced by a multitude of factors, including



• **Fig. 5.2** Anticipated timeline of progression from coping to acceptance.

personality, past mental health history and experience with adversity, support networks, and access to resources. Some patients may progress through these stages relatively smoothly, whereas others may experience prolonged periods of stagnation or regression. This process of acceptance is not necessarily linear, with patients often cycling through various stages multiple times as they confront new challenges or milestones in their journey.

Upon first learning of their injury, it is not uncommon for individuals to struggle to accept the reality of their situation. This phase may be characterized by feelings of shock, numbness, disbelief, and loss. At first, the focus is simply coping with the circumstances that are the challenges of the day: managing pain, basic care needs, the complexities of the health care system; gaining a basic understanding of the illness or disability; and establishing relationships with their providers. Denial may be a component of coping because it makes what is otherwise overwhelming more manageable. As individuals establish a degree of mastery over these basics, denial gives way to a more active engagement with the challenges at hand. Negotiating this initial stage is crucial because it sets the foundation for subsequent phases of adjustment.

On transitioning from acute care to the rehabilitation setting, patients and their families move from the immediacy of medical intervention toward the longer term journey of recovery. They encounter a series of challenges and considerations that shape their adjustment trajectory. As the focus of care moves toward stabilization and preparation for returning to the community, the impact of the injury and its implications for the future become more apparent. Immediate evaluations by therapy professionals help patients understand the full impact of their injuries, which can be disorienting and emotionally overwhelming, as patients confront the reality of their condition and the uncertainties of

what lies ahead and grapple with the contrast between their pre-injury abilities and current limitations. How will they be able to participate in essential roles and activities at home, at work, and in the community? Questions related to sexuality and intimate relationships, reactions to their condition, financial concerns, future-related questions, support systems, and living situations are likely. Individuals may see their disability as a threat to their identity, such as the runner who is no longer able to walk or the lecturer who is now aphasic. They may question their purpose, value, or worth. They may express concerns about being a burden on those around them. Emotions related to dignity may arise as patients confront the loss of bodily autonomy and the need for assistance with basic activities. A lack of education about the potential for rehabilitation and recovery can worsen feelings of hopelessness. Disability bias can make it difficult for patients and their families to envision a quality future, leading to withdrawal, reluctance to seek help, and expressions of emotional distress, further isolating them from available support. The etiology of the injury or illness can impact their reactions, whether it was caused by an unexpected event such as a stroke, the actions of others, or a result of their own decisions or actions.

Transition Through Rehabilitation Settings

As individuals with injuries or disabilities approach discharge from medical or rehabilitation settings, they often confront many fears and uncertainties. The fear of not being ready for discharge is a common apprehension among patients. They may not have achieved the hoped-for level of recovery and are worried about their ability to manage daily tasks or cope with challenges outside of the supportive rehabilitation environment. This fear is often compounded by concerns about the lack of around-the-clock support, as individuals confront the reality of transitioning from a setting where assistance and expertise are readily available to one where they must be more reliant on themselves and their support networks. This anxiety is likely compounded for patients whose injuries are a result of an unexpected medical event, such as a stroke or heart attack, or had a complicated medical course with setbacks or return to acute care. Family members supporting the individual, who may have been traumatized by the injury events and early hospitalization, may express their reluctance to assume responsibility for someone they see as vulnerable and still medically unstable. Fear of isolation can also loom large as individuals contemplate life outside of the rehabilitation facility. The prospect of returning to familiar, albeit less accessible environments can be daunting, particularly if individuals struggle to envision themselves navigating these spaces with their new limitations. Additionally, the resurgence of preexisting problems, whether physical, emotional, or social, can further exacerbate anxieties and doubts about the future. Other factors influencing adjustment include chronologic age, financial resources, spiritual beliefs, and access to support services. Employment significantly enhances self-esteem and psychological adjustment for individuals with disabilities.¹²

During the first year after injury, it is common for patients to experience a variety of emotional or psychologically reactive symptoms related to change and loss.⁶⁹ They can be intense and may include sadness, grief, anger, frustration, guilt, confusion, fear, and feelings of being overwhelmed, as individuals struggle to make sense of their new reality. These are natural responses to the perceived threat to one's physical and psychological sense of self and well-being. Sadness, depression, and anxiety may become more consistent, as individuals mourn the loss of their former

selves and struggle with the challenges of navigating an uncertain future. Anger and acting-out behaviors may present as expressions of frustration and powerlessness in the face of adversity, while feelings of helplessness, inadequacy, and vulnerability may take a toll on self-esteem and confidence.⁶⁵ Feelings of hopelessness and helplessness may trigger suicidal ideation. Individuals and their families are likely to go through a period of grieving during these early stages in response to the multiple losses associated with illness and acquired disability as they mourn the loss of what was and what was expected to be. There are several descriptive models describing grief as a progression through a series of defined stages or emotional phases as people grieve their losses.⁴² These earlier theories proposed that the emotional responses were experienced in a stagelike manner ending with acceptance of the loss. Currently, however, it is recognized that grief is an individualized process that does not necessarily progress through a series of stages. It has been described as an "oscillation between loss-orientation, marked by intrusive, painful feelings of loss and thoughts of what used to be, and restoration-orientation, where one has more hopeful feelings and a focus on adjustment to loss."³⁸ It may wax and wane over the course of an individual's recovery and can be triggered in response to encountering barriers or reminders of loss, such as injury anniversaries, altered life events, or milestones in which the individual is not able to participate as expected.¹³ It can be influenced by social support and cultural influences from country of origin, religion, or spirituality.^{64,70} Providers can support patients through the grieving process by checking for grief, listening sympathetically, and talking about and normalizing the grieving process, seeing the patient as a whole to appreciate the complex biopsychosocial impact of the illness or disability.³⁸

Mental Health Risks

People living with chronic medical conditions and disability are at risk for problems with mental health and substance misuse. The Centers for Disease Control and Prevention published a report in 2020 indicating that adults with disabilities report experiencing more mental distress than those without disabilities.¹⁵ This is supported by multiple studies examining the mental health consequences of a variety of disabilities. Relative to the general population, higher rates of depression, anxiety, and posttraumatic stress disorder (PTSD) have been found in survivors of traumatic brain injury,^{29,31-23} SCI,^{10,65} limb loss,^{3,8,17,56,59} burns,⁶³ and stroke.⁴⁵ People with disabilities are at 2.5 times higher risk for suicide than people without disability with the risk increasing with the number of disabilities.⁴⁴ Rates of substance abuse have been found to be higher in individuals with a disability than those without.²⁶ This increased risk for mental health disorders has been attributed to the biologic (pain, sleep, incontinence, medication effects, damage to emotional regulatory mechanisms, etc.), social (loss of social supports and subsequent isolation, social role disruption, family discord, financial strains, housing insecurity, ableism, social stigma, unemployment), and psychological (loss of independence and autonomy, loss of identity and sense of self, exclusion, shame, sexual health, uncertainty about the future, loss of control) changes associated with the disability.^{12,34,37,65} How the injury occurred, the severity and visibility of the disability, death or injury of another associated with the injury, past history of trauma, socioeconomic level, level of education, litigation, potential for recovery, stability of the disability, prior personality, and coping strategies can impact individuals' ability to cope and adapt.^{63,65} Cognitive abilities may be compromised because of

acquired brain injury, delirium, medication effects, sleep deprivation, pain, or mood disorders. Problems with information processing and retention, receptive and expressive communication, problem solving, and emotional regulation complicate adjustment and adaptation.⁶⁵

Poor adjustment can take a variety of forms, negatively impacting rehabilitation and quality of life. Treatment refusal, disengagement, noncompliance, and premature withdrawal from treatment may be indications of a struggle to adapt to changed circumstances. Social withdrawal exacerbates loneliness and emotional distress. Unhealthy coping behaviors, such as substance abuse and self-harm, may offer temporary relief but worsen long-term adjustment,¹² impacting rehabilitation and quality of life. Untreated psychological distress, including anxiety and repression, often accompany maladjustment and can significantly hinder rehabilitation outcomes and overall quality of life by reducing functional gains, increasing follow-up care needs, and exacerbating medical comorbidities. Depression and anxiety significantly contribute to the likelihood of rehospitalization and death within the first 6 months of stroke.¹⁹

For individuals with lower-limb amputations, social discomfort and body-image anxiety are associated with poorer adjustment, leading to greater activity restriction, depression, and anxiety. Nonadherence to treatment can result in poorer psychosocial recovery and higher health care costs in this population.³⁰ Physiatrists play a key role in this process, often encountering patients with suicidal behaviors and forming meaningful therapeutic alliances. They are frequently the first point of contact for patients seeking treatment for physical ailments that may be masking mental health symptoms. Extended treatment periods allow physiatrists to identify and address these concealed symptoms, crucial for successful rehabilitation.³⁷ Recognizing and addressing these indicators of maladjustment is essential. Health care providers and support networks should intervene early, offering mental health services, social support, and resources for coping and resilience. This proactive approach can help individuals navigate challenges more effectively, promoting greater well-being and quality of life.^{25,52}

The majority of patients discharge home to their families in the community after their rehabilitation. While much of the focus of care is on the individual, it is important to consider the impact of the injury or disability on the individual's family who are most typically the primary source of care and support. Disability and chronic illness impact the family system in much the same way as the individual. Families face increased physical, emotional, social, and financial stressors caused by decreased resources, the increased time and energy demands of care, and social isolation. Demands of care may exceed caregiver's physical and emotional capacities.^{5,74} Families may bear the brunt of the survivor's struggles with adjustment.⁵⁰ This is likely to be further exacerbated in families facing changes in cognitive abilities and emotional and behavioral dysregulation.⁷⁵

Families may go through a grieving process with emotional reactions similar to the individual's, which may be complicated by feelings of guilt or shame if they are not living up to expectations.^{5,28,33,41} Caregivers are at risk for fatigue and burnout resulting in poor emotional and physical health as well as decreased well-being and quality of life.^{5,16,21,41,74} Couples can experience multiple losses: a sense of security, plans for the future, a sense of connection, physical and emotional intimacy, and mutual support.^{21,28} The family's ability to cope and adjust is influenced in much the same way as the patient's by the presence or absence of emotional, financial, social,

and spiritual resources; mental health history; history of trauma, conflict, or adversity; and reliance on positive or negative coping strategies. Support for family coping and adjustment is essential as multiple studies have established the link between family adjustment and patient outcome.^{2,5,28,33,75} Family needs for support can be grouped into categories: information (about the illness/injury/disability, rehabilitation process, prognosis, resource availability), emotional (validation of feelings, inclusion in the care process, social connection), instrumental (identifying available benefits, community supports and resources, equipment acquisition, environmental modification), and training (disability/disease management, care needs, self-care, and positive coping strategies).¹⁴ Support in managing the disruptions to normal family functioning, training in problem-solving techniques, identifying and negotiating caregiver needs, along with access to group or peer-to-peer supports to reduce isolation is recommended for caregivers.^{41,50,74} Family counseling and peer support groups can facilitate communication and foster resilience within relationships.⁶⁷

Fortunately, most survivors adapt to their new circumstances and achieve an acceptable level of quality of life. Adaptation after SCI, as an example, is a natural process that involves both internal and external elements. Resilience and positive adaptation are common among people with SCI with about 60% reporting a meaningful and fulfilling life after injury. Internally, individuals may experience grief, loss, uncertainty, changes in thinking, and shifts in identity and self-efficacy. Externally, adaptation involves learning to address structural and systemic barriers to navigate environmental and social spaces with a disability. Although significant lifestyle changes often occur within the first few years postinjury, adaptation continues throughout one's life and varies based on culture, situation, and context.

Factors Impacting Adjustment and Resilience

Factors that support resilience and positive adaptation include optimism, positive self-esteem, strong support systems, problem-solving skills, positive thinking, hope, perseverance, determination, self-efficacy, a sense of purpose, and spirituality.⁶⁵ As individuals continue to adjust, they integrate their disability into their sense of self-identity. They develop coping strategies, acquire new skills, and seek out resources to enhance their quality of life. This middle phase involves a gradual reconstruction of life goals, priorities, and aspirations incorporating disability limitations, remaining strengths, and new skills. In the late or long-term phase, acceptance gradually emerges as individuals reconcile with their new reality and begin to find meaning and purpose amidst adversity. This phase is marked by accommodation and adaptation, as individuals develop coping strategies, harness their strengths, find new expressions of traditional values and core principles, and build resilience in the face of ongoing challenges. Although acceptance may signify a sense of peace and equilibrium, it does not preclude the possibility of emotional upheaval or setbacks. With time and support, individuals begin to accept the reality of their disability and adapt to its challenges. This phase involves redefining one's identity and capabilities, finding new ways to navigate the world and pursue meaningful activities. Acceptance is not resignation to one's condition or situation but a proactive stance toward living with a disability and connecting with remaining strengths and assets on which to build recovery.^{22,51} Achieving a balance between acknowledging reality and nurturing hope is a delicate yet essential aspect of successful

adjustment. Although it is important to confront the challenges and limitations imposed by injury or disability, maintaining a sense of hope and optimism can fuel motivation, resilience, and perseverance in the face of adversity.

Resilience and posttraumatic growth are two important concepts that help to explain this postinjury recovery and successful adjustment. Resilience has variously been defined as the “process of, capacity for, or outcome of successful adaptation despite challenging or threatening circumstances,” “a dynamic process encompassing positive adaptation within the context of significant adversity,” and “the capacity of individuals to cope successfully with significant change, adversity or risk.”²⁴ Resilience has been associated with positive thinking and positive emotions, an active coping style, effective emotional self-regulation, psychological strength, adaptive coping, and favorable rehabilitation outcomes.⁵⁸ These definitions suggest a response to adversity that takes place over time as individuals initially cope with and then adapt to challenging circumstances. Their capacity for resilience is influenced by a variety of genetic, epigenetic, developmental, psychological, and neurochemical factors; past experience; internal and external resources, such as locus of control; and social, emotional, and financial supports.⁷³ Posttraumatic growth (PTG) was first identified as “the experience of positive change that occurs as a result of the struggle with highly challenging life crises” leading to “an increased appreciation of life, more meaningful interpersonal relationships, increased sense of personal strength, changed priorities, and a richer existential and spiritual life.”⁴ Resilient individuals tend to have a positive outlook on life as they engage in active problem solving, maintain an internal locus of control and emotional connections with others, and remain optimistic about the future.²⁷ Both resilience and PTG can be influenced through interventions, such as cognitive behavioral therapy (CBT), mindfulness training, exercise, encouragement of active coping strategies, promotion of humor, optimism and social connection, connecting with purpose, and personal values.^{51,73} Psychological factors, such as self-efficacy, mastery, purpose in life, and optimism, along with social factors, such as social support and social skills, are consistently associated with better mental health in individuals with SCI. These predictors of mental health are similar to those found in the general population and in individuals with other chronic health conditions. Mental health problems are not inevitable, and the interplay of psychological and social factors plays a crucial role in both the occurrence and the persistence of these issues.⁶⁸

Individuals employ a variety of strategies to manage distress and maintain psychological equilibrium. These coping behaviors can be classified into two broad categories: approach-oriented coping and avoidance-oriented coping. Approach-oriented coping strategies, such as using problem-focused coping, seeking information, reframing perspectives, and engaging in relaxation techniques, enhance resilience and facilitate emotional regulation and adjustment. Active problem-focused coping involves seeking solutions to practical challenges, whereas emotion-focused coping addresses emotional distress and helps find meaning in experiences.⁷³ Flexible, adaptive coping styles are generally associated with better psychological adjustment over time. Persson and Ryden describe coping as a “multidimensional, fluctuating process.”⁵³ Effective coping involves a variety of psychologically based strategies that are used in different situations. These include:

- **Problem-reducing actions:** active efforts to reduce or circumvent the problems created by or associated with the disability
- **Change of values:** conducting an existential review, a reevaluation of life and what is truly important, finding meaning

within the disability, identifying interests less in conflict with the disability and more in line with remaining strengths

- **Establishing social trust:** having close and supportive relationships, identifying people on whom one can rely, asking for help when needed, meeting new people via support or community groups
- **Minimization:** playing down the personal threat posed by the disability: “It could have been worse” or “Others have it worse”
- **Self-trust:** belief in one’s capacity to master challenges

Effective coping involves finding a balance between the endpoints of two dimensions: acknowledgment of reality vs. creation of hope and trust in oneself vs. trust in others. This balance point may change depending on evolving demands or changes in circumstances, learning to control what one can and letting go of what one cannot, and being flexible in managing the disability and not extreme in any direction.⁵⁴ Importantly, these concepts and movement on these dimensions are subject to influence by role models, therapy, coaching, education, and goal setting. In contrast, more avoidance-oriented strategies (withdrawal, denial, or distraction; substance use; behavioral disengagement) may provide temporary relief from stress and adversity but hinder rehabilitation efforts.⁷³ They can serve as initial coping mechanisms, allowing individuals to process overwhelming emotions at their own pace. However, reliance on avoidance-oriented coping is linked to poorer adjustment, especially in the first year, as it hinders emotional processing, engagement in care, support seeking, acceptance, and growth.⁷¹ Expressions of intense emotions are not immediate indicators of maladjustment or psychopathology and should be acknowledged and validated by health care providers.³⁰ The use of different coping strategies can be expected to fluctuate over the course of recovery. Another way of measuring successful adjustment is psychological well-being. It is tied to a sense of optimism regarding the future, the belief that things can get better, often in response to indications of improved physical functioning. Psychological well-being has been associated with improved psychosocial adjustment, ability to cope with stress, and decreased depression and anxiety.⁵¹ Ryff and Keyes describe psychological well-being as characterized by the presence of certain attitudes and behaviors more than the absence of psychopathology.⁶¹ These include:

- **Self-acceptance:** being able to see both one’s strengths and weaknesses, having a sense of purpose, having goals and objectives that give life meaning and direction
- **Personal growth:** having a sense of personal talents and potential being realized; having positive relations with others, including close, valued connections with others
- **Environmental mastery:** being able to manage the demands of everyday life
- **Autonomy:** having the strength to follow personal convictions, having an internal locus of control, being motivated by self-determination

These qualities, attitudes, and skills can be encouraged or taught and can be a focus of rehabilitation interventions and goal setting.

At the onset, more avoidant strategies, such as denial and withdrawal, may be used to bring the volume of stressors to tolerable levels. At the same time, with education, encouragement in therapy, effective pain and sleep management, and preliminary mastery, the number of stressors reduces to more tolerable levels. With emerging mastery, time, and emotional support, cognitive appraisal changes so that the stressors themselves are seen as more tolerable, allowing for more active strategies. As individuals begin

to see indications of progress and mastery over aspects of their experience, hopes and expectations become more realistic. Goals shift, for instance, from complete restoration to maximizing participation in essential activities.⁵³ In interviews with stroke survivors, Basseotto et al. identified the importance of value-directed living (VDL; i.e., behaving in ways that reflect one's values).⁴ Injury or disability may require a review of values and priorities and exploring new ways of expressing those values. Greater engagement in VDL has been associated with increased psychological well-being, decreased distress, and PTG.⁴ With time and support, overwhelming emotions diminish in frequency and intensity as individuals adapt to their new reality and can begin to engage in more active coping strategies.^{12,15,65,73}

Assessing Psychological Health

Identifying risk factors for psychological maladjustment is essential for implementing preventive and rehabilitative strategies. Monitoring mood and psychological distress is vital, as untreated conditions can impact pain levels, goal achievement, and overall life satisfaction. A review of medical and psychiatric history, including trauma history, work and education history, family history, and social determinants of health (including social, financial, and spiritual resources), are part of any standard intake or history and physical. This helps paint a picture of the patient as a person and begin to identify strengths and weaknesses that may support or limit the adjustment/coping process. Asking about past experiences with challenge and adversity can give insight into resilience and coping style. As mentioned earlier, individuals with cognitive and physical disabilities are at higher risk for developing psychiatric and substance use disorders.¹⁵ Because these disorders can emerge at any point, psychological health and well-being should be assessed as part of a routine medical review. The Patient Health Questionnaire (PHQ-9) or Beck Depression Inventory are used frequently to screen for depression. The Generalized Anxiety Disorder (GAD-7) is recommended for screening for anxiety and the PTSD Checklist for PTSD.⁶⁵ The PHQ-9³⁶ and the GAD-7⁶⁶ are surveys that can be administered in a few minutes. Individuals are asked to identify the frequency with which symptoms of depression (PHQ-9) and anxiety (GAD-7) have been present in the past 2 weeks as well as the extent to which identified symptoms interfere with their ability to work, manage things at home, or get along with others. Responses result in a score that can then be used to rate level of severity and provide comparative data to monitor mood across sessions. For some individuals, it can be helpful to ask family members to respond to the questions to corroborate patient self-reports. Suicide risk increases with the number of disabilities, the presence of depression, a sense of burdensomeness, isolation, increased pain tolerance, and decreased fear of death.⁴⁴ Screening for suicidal ideation as part of routine encounters is recommended because this has been found to be effective in decreasing actual suicide attempts and follow-up with mental health care.⁴³ The Columbia Suicide Severity Rating Scale⁵⁵ is a five-item questionnaire that can be used to identify the presence and intensity of suicidal ideation to help determine level of risk and need for intervention either as a general screening tool or in response to indications of depression. Research has identified biopsychosocial risks for suicide in these populations, including worsening functional limitations, new illnesses, chronic pain, suicidal thoughts, prior attempts, mood or substance abuse disorders, hopelessness, personality disorders, and prior psychiatric hospitalizations.¹² There are a number of substance abuse screening

tools that can be used to identify problematic substance use, such as the Alcohol Use Disorders Identification Test (AUDIT), the Brief Michigan Alcohol Screening Test, the CAGE assessment, the Drug Abuse Screening Test (DAST), the Substance Abuse Screening Inventory, and the Short Michigan Alcohol Screening Test.⁴⁷ The Screening, Brief Intervention and Referral to Treatment is a tool that was developed by the Massachusetts Department of Public Health Bureau of Substance Abuse Services. It uses the AUDIT-C and the DAST-10 to identify problematic substance use. It then provides guidance in the use of motivational interviewing techniques to provide education regarding the risks associated with the substance use followed by referral for treatment as appropriate.¹¹

A history of trauma is common in the general population. At the same time, patients may have had traumatic experiences in the course of acquiring their injuries or during their care, such as in the intensive care unit. These experiences can impact physical and mental health and interactions with caregivers. Aspects of hospitalization, medical care, examination, and treatment may trigger anxiety, reexperiencing phenomena, avoidance, noncompliance, or emotional or behavioral disturbances. Raja et al. recommends the adoption of trauma-informed care as a universal practice approach regardless of trauma history.⁵⁶ The approach involves routine use of universal trauma precautions where trauma history is unknown (patient-centered communication and care, a focus on increasing the patient's sense of control and reducing anxiety, providing an overview of what is expected, asking about a patient's priorities or concerns about the visit or procedure, offering choices). Where history of trauma is known and the patient is expressing anxiety during the visit or procedure, more trauma-specific approaches are recommended (asking what can be done to increase patient comfort, identify specific sources of anxiety and have a signal that the patient can use to identify distress, acknowledging concerns, and developing coping strategies the patient can use to manage discomfort). They recommend routine screening and referral for more trauma-focused care as needed along with monitoring caregivers for indications of vicarious traumatization and burnout.⁵⁷

These psychological health tools are not intended to diagnose mental health or substance use disorders so much as identify when referral to mental health specialists may be warranted. Health care providers and support networks should intervene early, offering mental health services, social support, and resources for coping and resilience. This proactive approach can help individuals navigate challenges more effectively, promoting greater well-being and quality of life.^{25,52} Regular screening for poor adjustment allows for targeted interventions aimed at improving adaptation and overall well-being. By addressing these factors holistically and promoting social support, mental health support, empowerment, and accessibility, health care providers can mitigate the risk of maladjustment and foster resilience, well-being, and autonomy among individuals facing injury or disability.⁶⁸ This comprehensive approach ensures the timely provision and continuous support, enhancing overall quality of life and psychological well-being. Providing access to mental health services and resources for social support, coping, and resilience-building allows individuals to navigate the challenges of injury and disability more effectively.

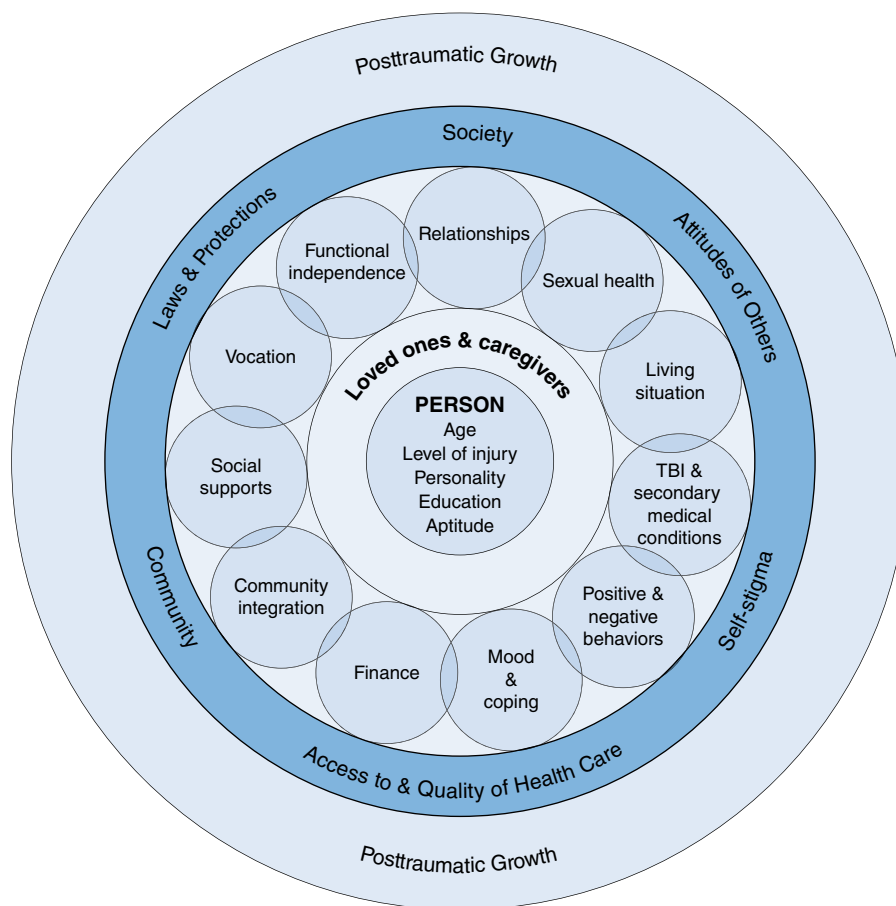
Psychological Interventions

Key approaches include CBT and mindfulness-based stress reduction (MBSR).^{7,9} CBT helps manage emotional distress and

enhances coping skills through techniques such as cognitive restructuring, problem solving, and assertiveness training. MBSR reduces stress and improves emotional regulation and resilience, complementing CBT's structured approach with mindful practices.⁴⁶ Patient motivation is a crucial factor in adapting to disability and managing related medical concerns. Motivational interviewing (MI) can help to foster patients' engagement by eliciting their motivations and guiding them toward healthy behavior changes.^{7,60} MI⁶¹ is an interventional approach initially developed by Miller and Rollnick in 1991 to encourage behavior change that came from the substance abuse community and has been used to influence change in a variety of behaviors.⁴⁸ MI is a relational style that involves empathic listening, expressing empathy for the patient's situation and potential ambivalence around change, developing discrepancy between patient's behaviors and goals for recovery, avoiding argumentation, rolling with resistance as a normal part of the change, and respecting and supporting the patient's capacity for change and self-efficacy.^{48,60} It engages the client in a conversation about behavior change, coming to an agreement on what behaviors to change, clarifying the client's motivation for changing, and generating realistic plans that support the client's confidence in being able to make the changes.⁶⁰ MI has been found to be more effective than traditional advice-giving with even brief contacts of 15 minutes with increased impact with more than one encounter.^{39,61}

Comprehensive Care Programs

A comprehensive understanding of psychological adjustment allows for tailored interventions that address the diverse needs of individuals navigating acceptance and healing. Recognizing the range of coping styles informs interventions that support individuals in building resilience and psychological well-being.¹² Actively involving patients and their families in decision-making processes ensures that care plans align with their preferences, values, and goals. Developing coordinated care plans that address physical, emotional, social, and vocational needs ensures that interventions from different providers complement each other. Continuity of care is also important to support smooth transitions between different care settings and maintaining consistent follow-up to monitor progress and adjust interventions as needed.⁴⁰ Availability and cost effectiveness of adaptive equipment, such as lightweight wheelchairs, communication devices, and home adaptations, enhance participation in daily activities. Ensuring autonomy in daily occupations through individual-centered adaptation and strong social networks is vital.¹ Helping to identify resources and opportunities for social participation, such as adaptive gyms and sports, accessible recreational facilities, peer support groups, and volunteer activities, supports social integration.³⁵ Fig. 5.3 reflects the multiple levels and layers beyond simply the individual impacted by injury and the process involved



• **Fig 5.3** Model illustrating the social and psychological impact of injury or disability. *TBI*, Traumatic brain injury. (From Budd MA, Gater Jr DR, Channell I. Psychosocial consequences of spinal cord injury: a narrative review. *J Pers Med*. 2022;12(7):1178.)

in working toward adaptation. Disability or illness brings about a range of biologic, psychological, and social consequences that collectively impact an individual's identity, well-being, and quality of life. Physically, it can cause impaired sensation, mobility issues, and increased health risks. Psychologically, it may lead to mood changes, altered self-image, and challenges in maintaining self-efficacy. Socially, it can affect financial stability, accessibility, and participation in community life, often exposing individuals to biases and discrimination. These factors combine to influence the individual's sense of purpose, values, and overall life satisfaction. Adjusting to such conditions is a dynamic process involving emotional turmoil, coping strategies, and a quest for regained control and meaning in life. Key factors include the nature and severity of the disability, preexisting mental health conditions, social support, financial strain, and personal coping styles. These elements can either facilitate or hinder adaptation, with positive coping strategies such as active problem solving and social support linked to better outcomes. Conversely, maladaptive strategies such as substance misuse and avoidance-coping contribute to psychological distress and poorer adjustment. Addressing these factors early on is critical in supporting individuals through the challenges of disability. Health care providers play a crucial role in supporting patients through this adjustment, considering such factors as personality traits, coping mechanisms, social support, and preexisting mental health conditions, which significantly impact how individuals adapt to adversity and maintain quality of life after injury or illness.

Addressing adjustment to disability in PM&R requires a comprehensive approach that includes thorough assessments, motivational interviewing, multidisciplinary collaboration, and a team-based care model. This collaboration may involve mental health professionals for counseling and medication management, occupational and physical therapists to develop rehabilitation plans, social workers and case managers to assist with services and resources, medical specialists for specific interventions, and nutritionists to create tailored dietary plans. Comprehensive assessments involve regular evaluations using various tools to gauge overall well-being and specific quality of life domains. Patient-centered interviews engage patients in discussions about their experiences and concerns, helping to identify personal values as areas needing intervention. Regular monitoring involves periodic reassessments to track changes in functional status, emotional well-being, and quality of life over time, allowing timely adjustment to care plans.¹⁸ The recovery process is dynamic, with individuals cycling through various stages multiple times as they encounter new challenges. Early intervention and support tailored to individual needs are essential in fostering positive adaptation postdisability. By focusing on these key areas, PM&R doctors can significantly enhance the quality of life and facilitate healthy adaptation for their patients and address their comprehensive needs. Integrated programs that provide comprehensive care and social support can reduce barriers and improve outcomes.^{40,72}

Summary

It is important to appreciate the meaning of the injury/disability by understanding that patients' priorities and values can assist with work in balancing hope with reality, building on remaining strengths, and setting short-term goals that align with long-term values and ambition. Allow for the chance to talk about life before illness/injury/disability to appreciate the injury's impact on essential roles and activities.

Comprehensive intervention strategies to employ include collaborative goal setting, which focuses on retained strengths, resources, and abilities, and encouraging a problem-solving stance involving information seeking and education. It is vital to encourage social support and involve family members, friends, and caregivers in education and support resources, and provide appropriate information, training, and support tailored to each phase of recovery.¹⁴ Implement strategies that address psychological, social, and physical needs. Identify resources both internal and external to support coping, adaptation, adjustment, and socialization. Utilize a variety of resources, including media (first-person narrative, YouTube, Ted Talks), faith or spiritual perspectives, adaptive sports, peer mentors, support groups, advocacy groups, and volunteering.²⁰

Routine assessment of coping, suicide risk, and need for psychological/emotional support is vital for individuals with high-risk factors.⁴³ Monitoring and support can in turn enhance engagement in rehabilitation.

CBT, MI, and cognitive reframing can help to assist patients in redefining priorities and finding alternative activities in line with remaining strengths, reinforcing a belief in one's ability to manage challenges and achieve goals through active engagement in their care and use of adaptive coping strategies such as problem solving and seeking social support.

Mental health consultation with psychology/psychiatry/social work can address prior psychiatric history interfering with adjustment, substance abuse, suicidal ideation, and problems with individual and family adjustment. Goals of care can include facilitating a (nonlinear) grieving process as patients mourn the loss of preinjury life and expectations, emotion-regulation training, relaxation, mindfulness, wellness training, and learning to control what can be controlled.

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